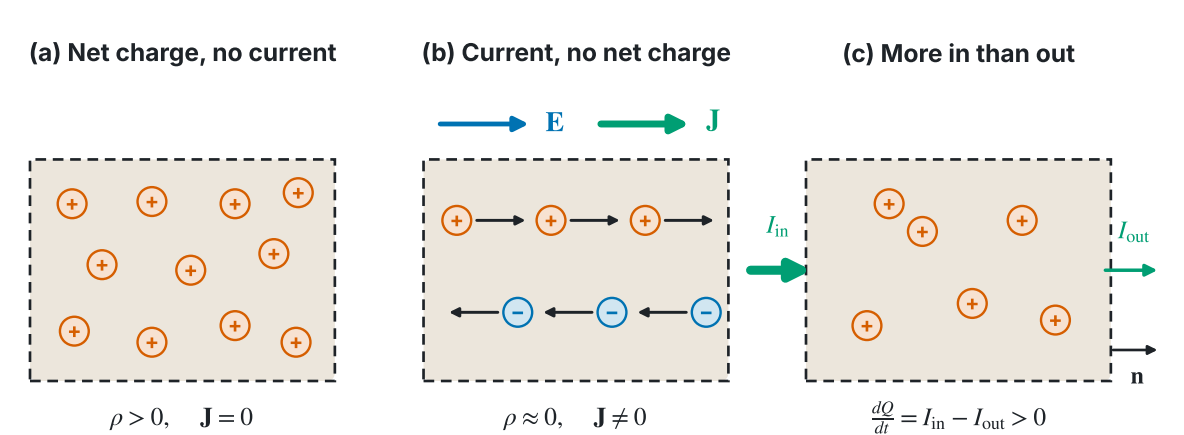
(a) Net charge, no current
(b) Current, no net charge
(c) More in than out
E
J
I_{in}
I_{out}
n
$\rho > 0, \quad \mathbf{J} = 0$
$\rho \approx 0, \quad \mathbf{J} \neq 0$
$\frac{dQ}{dt} = I_{\text{in}} - I_{\text{out}} > 0$
e.g. charge on a conductor at rest
both carriers add to J
dashed: a bookkeeping surface, not a wall